

Network Security Cyber laws & Compliance standards.

Explain working of Kerberos.

In a Kerberos system, there is a designated site on network, called Kerberos server which performs centralized key management & administrative functions.

The server maintains a database containing the secret keys of all users, generates session keys whenever two users wish to communicate securely & authenticates the identity of user who requests certain network services.

The Kerberos protocol uses strong cryptography so that a client can prove its identity to a server (and vice versa) across an insecure network connection.

The Kerberos authentication process uses same at both ends for each exchange. AS-REQ is the initial user authentication request. This message is directed to the KDC component known as Authentication Server (AS).

AS-RFP is reply of the Authentication server to previous request. Basically it contains the TGT (encrypted using the TGS secret key).

TGS-REQ is request from client to the Ticket Granting Server (TGS) for a service ticket.

TGS-RFP is reply of Ticket Granting server to previous request. It contains requested service ticket & service session key generated by TGS & encrypted using previous session key generated by the AS.

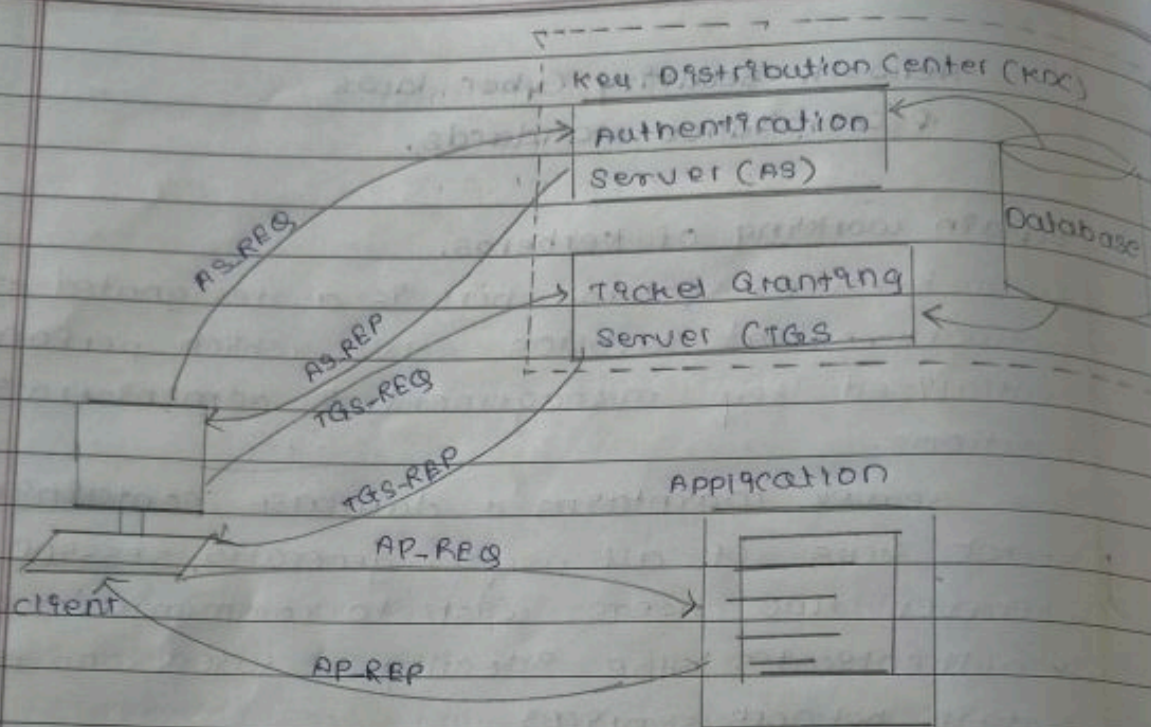


fig. working of kerberos

2. Explain Email-Security techniques, protocols.

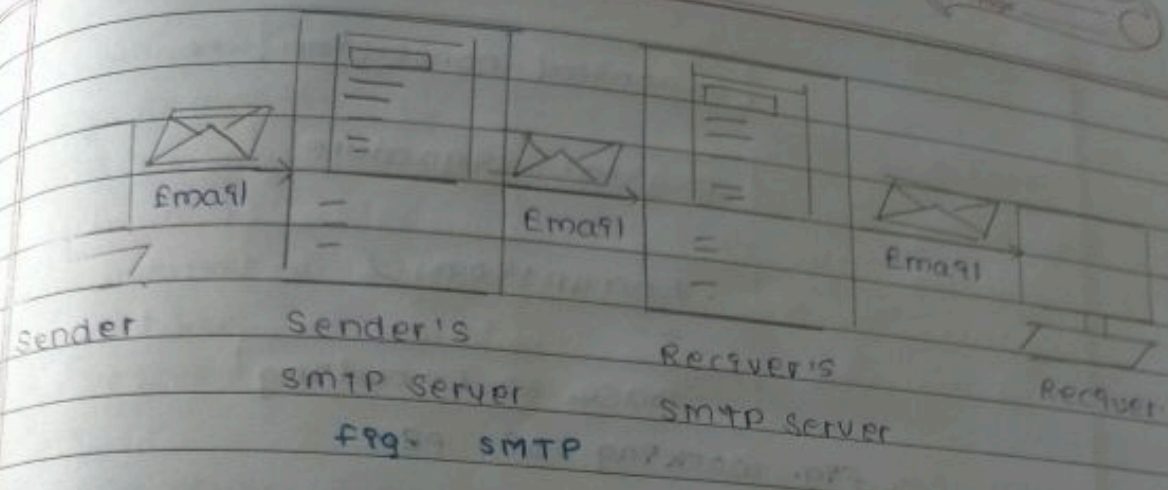
- 1) Simple mail Transfer Protocol (SMTP)
- 2) Privacy Enhanced mail (PEM)
- 3) Pretty Good Privacy (PGP)

1) SMTP

SMTP stands for Simple Mail Transfer Protocol. SMTP is a set of communication rules permit software to communicate an electronic mail over internet. It is called simple mail Transfer protocol.

SMTP also provides:

- send one message to one or many receivers.
- sending of any type of message is allowed such as text, audio, video etc.
- It can also send the message on different network outside the internet.



Advantages.

1. Not coding required
2. popular service
3. low cost
4. simple

Disadvantages.

- some firewalls can block ports commonly used with smtp
- Not support to audio & video.

2) PGM

Privacy-Enhanced mail (PEM) is an internet standard that provides for secure exchange of electronic mail. PEM does not require the use of a specific of algorithm.

It allows use of several algorithms for data encryption, key management & data integrity. Three cryptographic functions of PEM are.

1. Message integrity
2. Sender Authentication (Non Repudiation)
3. Encryption

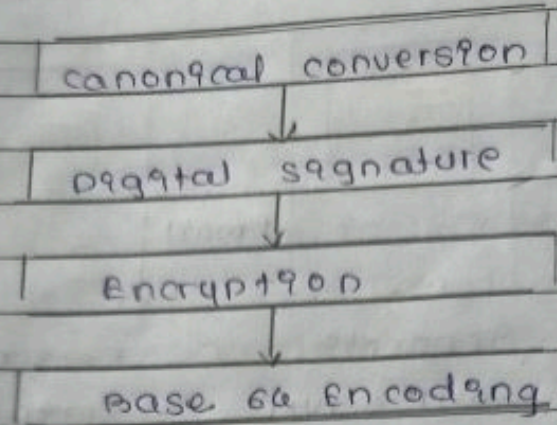


fig. working of PEM

3) PGP

PGP stands for Pretty Good Privacy (PGP) Protocol. It is invented by Phil Zimmermann. PGP was designed to provide all four aspects of security i.e. privacy, integrity, authentication & non-repudiation in sending of e-mail. PGP uses a digital signature (a combination of hashing & public key encryption) to provide integrity, authentication & non-repudiation.

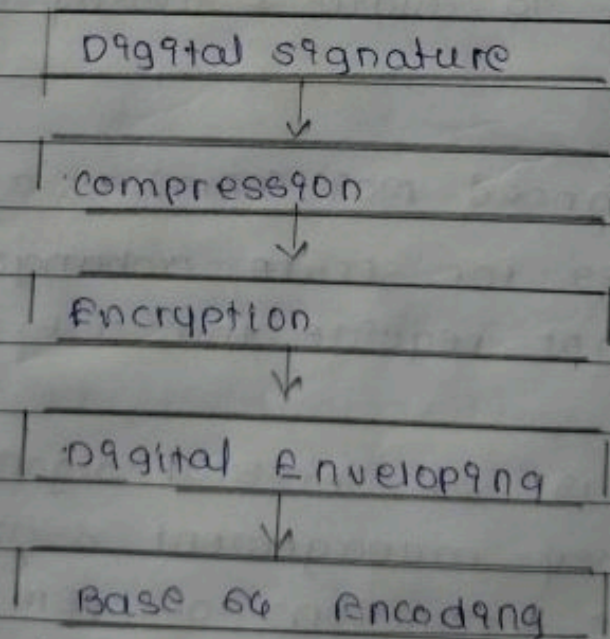


fig PGP Operation

Advantages of PGP:

- No compatibility problems
- It is free

Disadvantages of PGP:

- Complexity
- No recovery

Define Cyber Crime: List out Cyber Crimes.

Cyber Crime:-

- Cybercrime is a generic term that refers to all criminal activities done using the medium of computer, the internet cyberspace & the worldwide web (or web)

- Cybercrime (computer crime) is an illegal behavior directed by means of electronic operations that targets the security of computer system and data processed by them.

- Computer related crime is also called as "cybercrime" or "e-crime" or "digital technology crime". In other words the crime that involves & uses computer devices & internet, is known as cybercrime

List of cyber crimes.

1) Hacking

2) Theft

3) Cyberstalking

4) Identity Theft

5) Malicious software

6) Online scams

4. Define Hacking. Explain types of Hackers.
Hacking.

Hacking refers to unauthorized access of another computer system. It is the practice of modifying features of a system in order to accomplish a goal outside of the creator's original purpose.

Types of Hackers

1) **White Hat Hacker:-**

This type of hacker is someone who has non-malicious purpose whenever he breaks into a security system. It is also known as an Ethical Hacker.

2) **Black Hat Hacker:-**

This type of hacker is also known as a cracker & he has a malicious purpose whenever he goes about breaking into a computer security system with the use of n/w.

3) **Grey Hat Hacker:-**

This type of hacker is a combination of both white & black hat hackers. It is not a penetration tester but will go ahead & exploit a system if it is vulnerable.

4) **Elite Hacker:-**

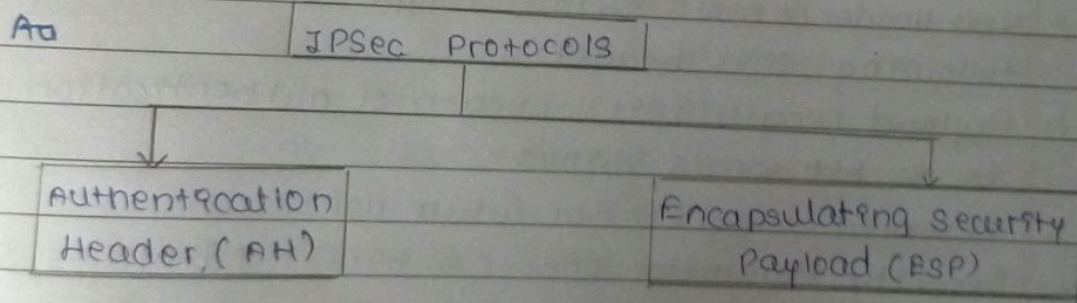
As with any society, better than average people are rewarded for their talent & treated as special. They are the masters of deception.

5) script kiddie:-

A script kiddie is a basically an part time or non-expert Hacker, who breaks into people's computer system. Underlying the concept of how those scripts he was on hand works.

5. What is IP Security? Explain AH & ESP Protocol in detail.

The IP Security (IPSec) is an Internet Engineering Task Force (IETF) standard suite of protocols involving two communication points across IP network which gives data authentication, integrity, and confidentiality. IPSec gives information related to the encryption, decryption and authenticated packets.



Authentication Header (AH)

AH protocol mainly focuses on authentication. The AH protocol is designed to provide data integrity & IP packet authentication. It also provides data integrity, data origin & an optional replay protection service.

This protocol provides authentication of all or part of contents of a packets by addition of a header. The IPSec AH is a header in IP packet which contains cryptographic checksum for contents of packet.

security is completely present in the contents of structure for AH is shown

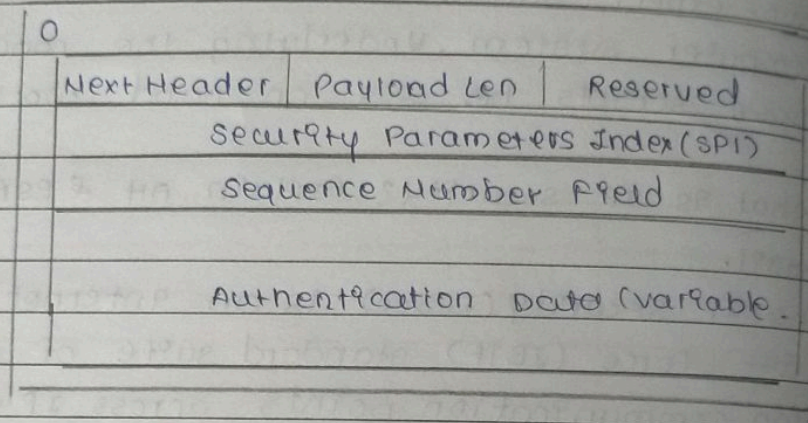


Fig. Authentication Header (AH) Format

Authentication header consist following fields:

1. Next Header (8 Bits): Identifies type of header immediately following this header.
2. Payload length (8 Bits): Length of authentication header in 32 bit words minus 2.
3. Reserved (16 Bits): For future use.
4. Security Parameter Index (32 bits): Identifies Security Association (SA).
5. Sequence Number (32 bits): Monotonically increasing counter value.
6. Authentication Data (Variable size): A variable length field that contain Integrity checked value (ICV).

Encapsulation Security Protocol (ESP)

This protocol gives data confidentiality. ESP also offers security services such as confidentiality, integrity, origin authentication & optional replay resistance.

The services provided depends on options selected at the time of security association (SA) establishment. ESP protocol also defines new header to be inserted in packet.

In ESP, algorithm used for encryption & generating authenticator are determined by the attributes used to create the SA.

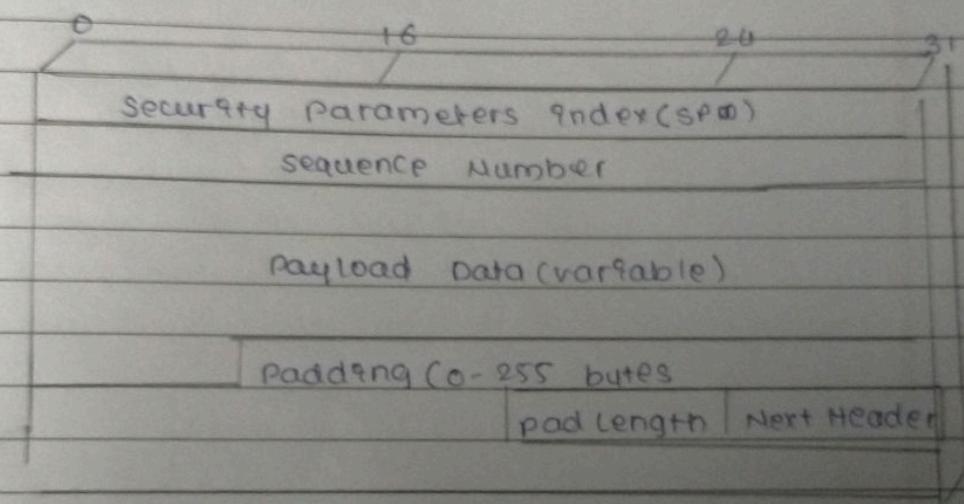


fig. ESP format

6. Explain the following terms.